A12 / ELECTRICAL DRAFT — NOT A MANUFACTURING RELEASE

All decoding/DSP stays on Raspberry Pi; processed output goes to amplifier by USB.

ARC candidate + four-input selector + LVDS. Routing prototype; DD+ and firmware not qualified.

power-control

File: power-control.kicad_sch

spdif

File: spdif.kicad_sch

selection-lvds

File: selection-lvds.kicad_sch

speakers

File: speakers.kicad_sch

arc-receiver

File: arc-receiver.kicad_sch

arc-connector

File: arc-connector.kicad_sch

arc-power

File: arc-power.kicad_sch

arc-decoupling

File: arc-decoupling.kicad_sch

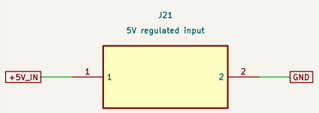
See requirements.md, sourcing.md and verification.md before using the model.

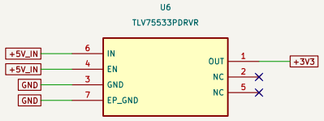
Sheet: /
File: rear-io-A12.kicad_sch

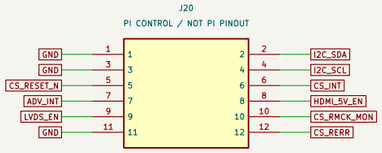
Title: Open Cinema rear I/O electrical study

Size: A3	Date: 2026-11-04	**Rev: A12 DRAFT**
KiCad E.D.A. 10.0.5		Id: 1/9

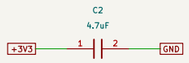
A12 ARC STUDY — 5V power and Pi control



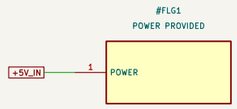


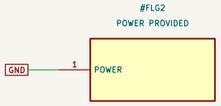






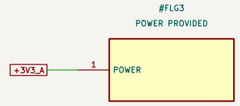
















5V regulated input only. U6 is now WSON-6 with grounded thermal pad.

R1/R2 DNP: use one I2C pull-up set at HAT; bus <=100 kHz. Both boards powered together.

J20 changed: pin7 ADV_INT, pin8 HDMI_5V_EN. Boot: CS reset low, LVDS and HDMI5V off.

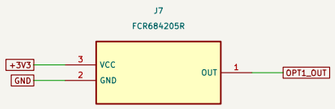
I2C: CS8416 0x20, ADV7511 0x39 (7-bit). Auxiliary ADV maps must not collide.

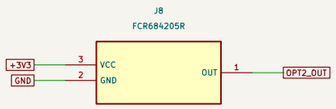
Sheet: /power-control/
File: power-control.kicad_sch

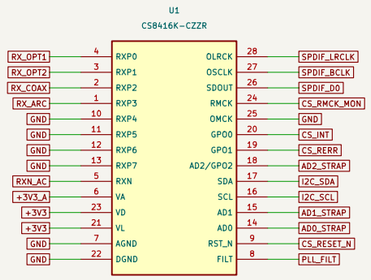
Title: Open Cinema — power-control

Size: A3	Date: 2026-11-04	**Rev: A12 DRAFT**
KiCad E.D.A. 10.0.5		Id: 2/9

A12 ARC STUDY — optical/coax/ARC shared selector













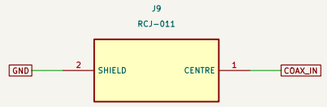








































RXP0 OPT1 / RXP1 OPT2 / RXP2 coax / RXP3 ARC. No resampling or codec decoding here.

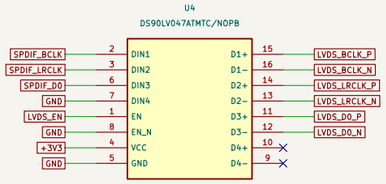
OMCK unused: carrier-rate measurement via RMCK needs Pi/HAT validation. D2 adds coax ESD clamping; system ESD testing remains required.

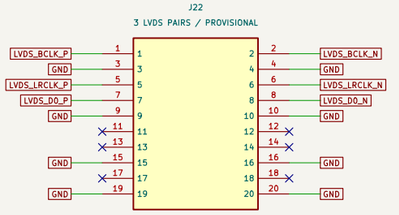
Sheet: /spdif/
File: spdif.kicad_sch

Title: Open Cinema — spdif

Size: A3	Date: 2026-11-04	**Rev: A12 DRAFT**
KiCad E.D.A. 10.0.5		Id: 3/9

A12 ARC STUDY — three active LVDS pairs







CS8416 owns source selection. A10 muxes, second transmitter and external eARC header removed.

LVDS carries BCLK / LRCLK / DATA0. J22 retains old pair positions; unused lanes are NC.

Pi is clock consumer. Receiver-side 100R termination and capture timing still need qualification.

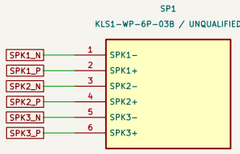
Disable LVDS and stop capture before changing source. Never play encoded words as PCM.

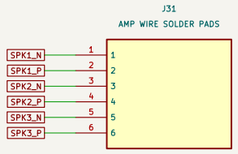
Sheet: /selection-lvds/
File: selection-lvds.kicad_sch

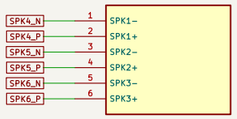
Title: Open Cinema — selection-lvds

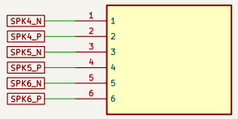
Size: A3	Date: 2026-11-04	**Rev: A12 DRAFT**
KiCad E.D.A. 10.0.5		Id: 4/9

A12 ARC STUDY — six independent speaker pairs









Six independent BTL pairs: speaker negatives must never share a return or connect to ground.

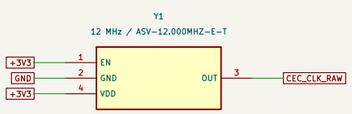
KLS clip current rating and exact supply remain unqualified. No fabrication/order release.

Sheet: /speakers/
File: speakers.kicad_sch

Title: Open Cinema — speakers

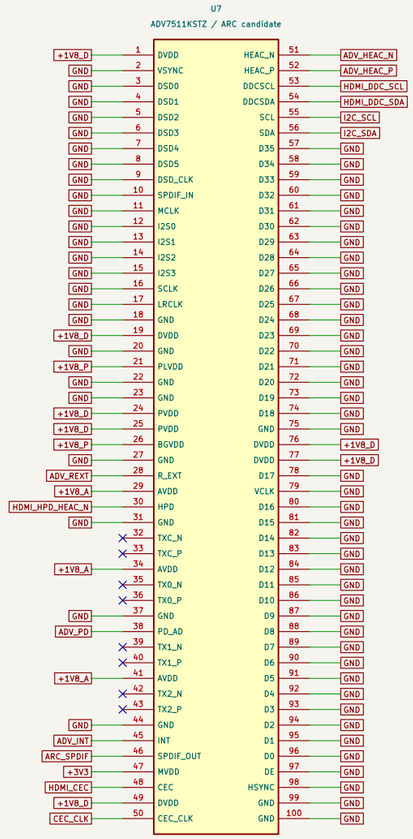
Size: A3	Date: 2026-11-04	**Rev: A12 DRAFT**
KiCad E.D.A. 10.0.5		Id: 5/9

A12 ARC STUDY — ADV7511KSTZ (100-pin LQFP)

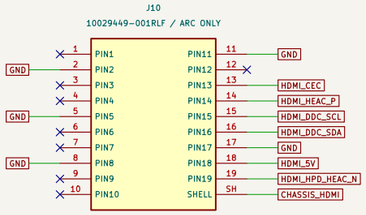







U7
ADV7511KSTZ / ARC candidate
+1V8_D
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
+1V8_D
GND
+1V8_P
GND
GND
+1V8_D
+1V8_D
+1V8_P
GND
ADV_REXT
+1V8_A
HDMI_HPD_HEAC_N
GND
+1V8_A
GND
ADV_PD
+1V8_A
GND
ADV_INT
ARC_SPDIF
+3V3
HDMI_CEC
+1V8_D
CEC_CLK
1 DVDD
2 VSYNC
3 DSD0
4 DSD1
5 DSD2
6 DSD3
7 DSD4
8 DSD5
9 DSD_CLK
10 SPDIF_IN
11 MCLK
12 I2S0
13 I2S1
14 I2S2
15 I2S3
16 SCLK
17 LRCLK
18 GND
19 DVDD
20 GND
21 PLVDD
22 GND
23 GND
24 PVDD
25 PVDD
26 BGVDD
27 GND
28 R_EXT
29 AVDD
30 HPD
31 GND
32 TXC_N
33 TXC_P
34 AVDD
35 TX0_N
36 TX0_P
37 GND
38 PD_AD
39 TX1_N
40 TX1_P
41 AVDD
42 TX2_N
43 TX2_P
44 GND
45 INT
46 SPDIF_OUT
47 MVDD
48 CEC
49 DVDD
50 CEC_CLK
51 HEAC_N
52 HEAC_P
53 DDCSCL
54 DDCSDA
55 SCL
56 SDA
57 D35
58 D34
59 D33
60 D32
61 D31
62 D30
63 D29
64 D28
65 D27
66 D26
67 D25
68 D24
69 D23
70 D22
71 D21
72 D20
73 D19
74 D18
75 GND
76 DVDD
77 DVDD
78 D17
79 VCLK
80 D16
81 D15
82 D14
83 D13
84 D12
85 D11
86 D10
87 D9
88 D8
89 D7
90 D6
91 D5
92 D4
93 D3
94 D2
95 D1
96 D0
97 DE
98 HSYNC
99 GND
100 GND
ADV_HEAC_N
ADV_HEAC_P
HDMI_DDC_SCL
HDMI_DDC_SDA
I2C_SCL
I2C_SDA
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
+1V8_D
+1V8_D
GND
GND
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GND
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GND
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GND
GND
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GND
GND
GND
GND

A12 ARC STUDY — HDMI coupling, CEC, DDC and protection









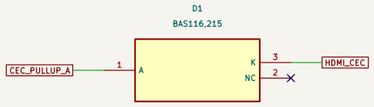








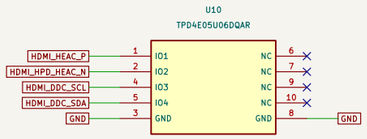


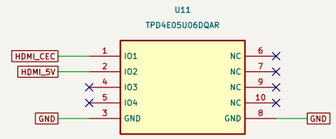












HEAC bias follows HW guide Fig17; eval variants differ. Verify bias, ESD loading and HPD on bench.

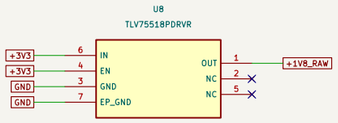
Connector-side ESD is provisioned; system ESD/CEC leakage, shell bond and back-powering not qualified.

Sheet: /arc-connector/
File: arc-connector.kicad_sch

Title: Open Cinema — arc-connector

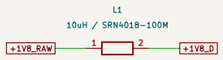
Size: A3	Date: 2026-11-04	**Rev: A12 DRAFT**
KiCad E.D.A. 10.0.5		Id: 7/9

A12 ARC STUDY — 1.8V supply filters and HDMI +5V

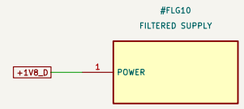


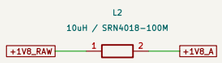




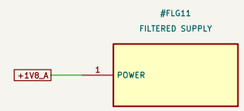






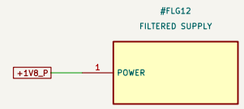


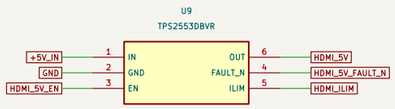












U9 ILIM=226k: approximately 101..142mA limit incl resistor tolerance (TI table). Not 75mA strap.

Sheet: /arc-power/
File: arc-power.kicad_sch

Title: Open Cinema — arc-power

Size: A3	Date: 2026-11-04	**Rev: A12 DRAFT**
KiCad E.D.A. 10.0.5		Id: 8/9

A12 ARC STUDY — local decoupling and HDMI power control







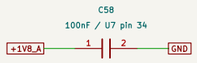














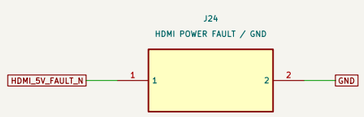


















Placement is a space study. Route decouplers at their named U7 pins with short ground vias.

Sheet: /arc-decoupling/	
File: arc-decoupling.kicad_sch	
Title: Open Cinema — arc-decoupling	
Size: A3	Date: 2026-11-04
KiCad E.D.A. 10.0.5	**Rev: A12 DRAFT**
	Id: 9/9